

- 4 Sickle cell anaemia is a non-infectious chronic disease. If not treated, sickle cell anaemia can be painful and life-threatening.

Sickle cell anaemia is caused by a base substitution mutation in the gene coding for the β -globin polypeptide of haemoglobin. This leads to a change in the primary structure of the polypeptide, as valine is present instead of glutamine. This results in abnormal sickle-shaped red blood cells, which stick together in blood vessels.

Symptoms of sickle cell anaemia include painful attacks when red blood cells block capillaries in tissues and organs.

- (a) Suggest the consequences to cells when sickle-shaped red blood cells block capillaries in tissues and organs.

.....

.....

.....

..... [2]

- (b) Sickle cell anaemia is an autosomal recessive inherited disorder:

- allele Hb^A codes for the normal β -globin polypeptide
- allele Hb^S codes for the sickle-cell polypeptide.

People who are heterozygous ($Hb^A Hb^S$) have sickle cell trait (SCT). For a child to inherit sickle cell anaemia ($Hb^S Hb^S$), both parents must have SCT.

A genetic screening program is available for sickle cell anaemia and SCT:

- when a mother is screened and found to have SCT, the father is then screened
- if the mother becomes pregnant, the fetus is screened for both sickle cell anaemia and for SCT
- the test is done either by amniocentesis or by chorionic villus sampling, both of which carry a small risk of the pregnancy failing.

- (i) Outline **two** advantages of genetic screening for sickle cell anaemia and SCT.

.....

.....

.....

.....

.....

..... [2]

- (ii) To test for the presence of Hb^S, DNA is extracted and the polymerase chain reaction (PCR) is carried out with two specific primers.

One mutation to produce Hb^S is a base substitution in the sixth codon of the β -globin gene. The normal codon GAG changes to GTG. The normal-specific primer detects GAG whereas the mutant-specific primer detects GTG.

Explain:

- why primers are used in PCR
- how the use of two specific primers allows the amplification of the normal, sickle cell anaemia and SCT genotypes.

.....

.....

.....

.....

.....

..... [3]

- (iii) Gel electrophoresis is carried out on the products of the PCRs.

Fig. 4.1 includes the results for two individuals, **A** and **B**, tested for the sickle cell allele.

- Each lane has an 860 base pair (bp) band to indicate the test is valid.
- Lane 1 is a control lane with a 207bp band for an individual with known normal phenotype.
- Lane 2 is a control lane with a 207bp band for an individual with known sickle cell anaemia phenotype.
- Lanes 1, 3 and 5 contain DNA from the PCR that used normal-specific primer.
- Lanes 2, 4 and 6 contain DNA from the PCR that used mutant-specific primer.

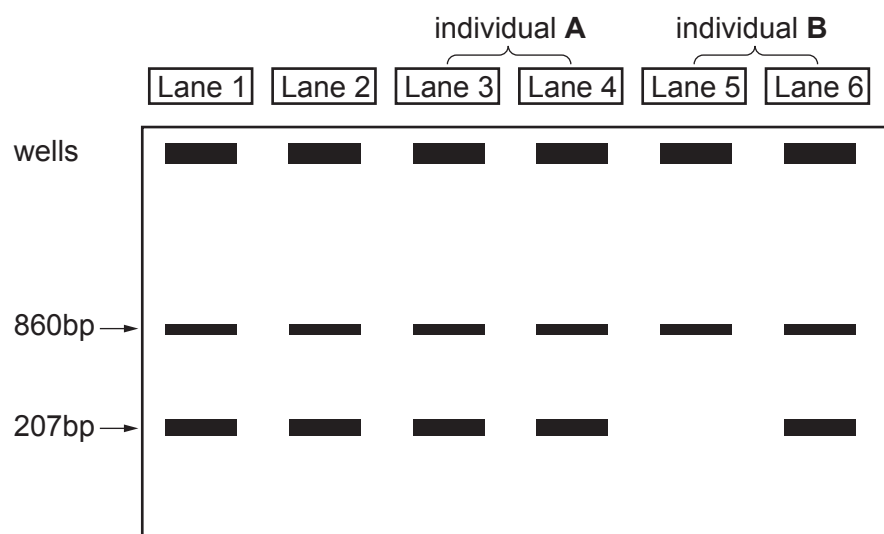


Fig. 4.1

Deduce the genotypes and phenotypes of individuals **A** and **B** in Fig. 4.1.

A

B

[2]

- (c) The number of cases of sickle cell anaemia is highest in sub-Saharan Africa, the Middle East and India. These areas also have a high incidence of malaria.

People with SCT (heterozygotes) are either unaffected or may have mild symptoms of sickle cell anaemia. One advantage of SCT is an increased resistance to malaria.

- (i) Explain how natural selection operates to maintain the presence of the sickle cell allele in populations in areas with malaria.

.....
.....
.....
.....
.....
.....
.....
.....
..... [4]

- (ii) Parents who use IVF to produce embryos may decide to have embryos genetically screened by a test known as pre-implantation genetic diagnosis (PGD). Only embryos that do not have sickle cell alleles are transferred to the woman's uterus.

Discuss **two** ethical reasons why parents using IVF may choose **not** to have PGD.

.....
.....
.....
.....
..... [2]

[Total: 15]